Exercises in Combinatorics

Section A

1. Initially 4 frogs are placed successively on a line, so that each frog is of distance 1 from the last one placed. In each move, a frog is allowed to choose any other frog as a center so that he can jump to the point symmetric with respect to the centre. Can it be possible, after finitely many moves, the 4 frogs are successively of distance 2008 units apart?

2. Two different numbers are written on each vertex of a convex 100-gon. Show that we may remove one number from each vertex, so that the two numbers of any two adjacent vertices are different.

3. Anna and Betty play the following game. First there are a pile of n coins. Anna first split them into two piles. Then Betty chooses a pile and again split it into two piles, and so on. The winner is the one who can split the coins into piles, so that each pile has at most two coins. Who has the winning strategy?

Section B

1. Let $n \in \{3900, 3901, ..., 3909\}$. Find those n which satisfy the condition: the set $\{1, 2, ..., n\}$ may be partitioned into different triplets (3-sets), so that in each triplet, one element is the sum of the other two.

2. Initially the edges of a complete graph with n vertices are colored by three different colors, with each color used at least once. Now assume suppose we choose at most k edges from this graph, and change the colors of these k edges to a particular color, so that the graph becomes connected using only the edges of this particular color. Determine the smallest possible value of k, so that no matter how the graph was initially colored, the goal may be attained.

3. Given a positive integer n, determine the maximum number of lattice points in the plane a square of side length $n + \frac{1}{2n+1}$ may cover.

4. Let $A_1, A_2, ..., A_n$ be finite sets satisfying

$$\left|A_i \cap A_{i+1}\right| > \frac{n-2}{n-1}\left|A_{i+1}\right|, i = 1, 2, \ldots, n \ (A_{n+1} = A_1).$$

Show that $\bigcap_{i=1}^{n} A_i \neq \Phi$.

5. We are given two square sheets of paper with area 2003. Suppose we divide each of these papers into 2003 polygons, each of area 1. (The divisions of the two pieces of these papers may be distinct. Then we place the two sheets of paper on top of each other. Show that we can place 2003 pins on the pieces of papers so that all 4006 polygons have been pieced.

Section C

1. Prove that: for any integer $n > 16$, there exists a set S consisting of n integers, with the following property: if S has a subset A, such that for any $a, a' \in A$, $a \neq a'$, $a + a' \notin S$, then $|A| \leq 4\sqrt{n}$.

27 August 2023

Section A

4. Initially 4 frogs are placed successively on a line, so that each frog is of distance 1 from the last one placed. In each move, a frog is allowed to choose any other frog as a center so that he can jump to the point symmetric with respect to the centre. Can it be possible, after finitely many moves, the 4 frogs are successively of distance 2008 units apart?

   (Chinese Western Mathematical Olympiad 2008). No, we may assume initially the frogs are placed on the points 1, 2, 3, and 4. Notice at any move (jump), a frog may only jump to a point of the same parity. Initially 2 frogs are at positions of odd parity and 2 at even parity. This cannot happen if the 4 frogs are of distance 2008 apart.

   Remark: Parity.

5. Two different numbers are written on each vertex of a convex 100-gon. Show that we may remove one number from each vertex, so that the two numbers of any two adjacent vertices are different.

   (Russian Mathematical Olympiad 2007). If each vertex has the same pair of number, say a and b, then we just have to keep a in the odd-numbered vertices and b in the even-numbered vertices. Suppose not, there exist a pair of adjacent vertices, A and B, so that A has the numbers a and b, and B has a different pair. Now number A as vertex 1, and B as vertex 100, and then number other vertices successively. Further we color the two numbers of A as red and blue, say a red, and b blue. Suppose we have color the numbers of the k^{th} vertex, say c and d by red and blue colors respectively, then for the $(k+1)^{\text{th}}$ vertex, if the vertex contains c or d (or both), then we just don't color c as red, nor d as blue, and so on. Then from vertex 1 onward, until vertex 100, no two adjacent vertices of the same color have the same number. Finally for the vertices B and A, again impossible both colors contain the same number. We then just keep a suitable color then.

   Remark: Presentation.

6. Anna and Betty play the following game. First there are a pile of n coins. Anna first split them into two piles. Then Betty chooses a pile and again split it into two piles, and so on. The winner is the one who can split the coins into piles, so that

each pile has at most two coins. Who has the winning strategy?

(Romania 2019). For $n=3,4,$ Anna wins in her first move. For $n \geq 5,$ and is odd, Anna has to split the pile into two, one with an odd number of coins, and the other with an even number of coins. Betty then spilt the pile into two, one with one coin, and the other pile an odd number of coins. So Anna has to split a pile with an odd number of coins into two, one with an odd number of coins, and another an even number of coins. Anna then applies the same strategy to split the even pile, and so on. At the end, Anna leaves piles of two kinds, one with 3 coins, and all other piles 1, and Betty can win. Or Anna eventually leaves one pile with 4 coins, and all other piles with 1, again And Betty can win. For $n \geq 6,$ and is even, Anna split it into a pile with 1 coin and other $n-1$ coins, and apply the strategy previously used by Betty, so Anna wins.

Remark: Parity analysis and strategy.

<u>Section B</u>

5. Let $n \in \{3900, 3901, ..., 3909\}$. Find those n which satisfy the condition: the set $\{1, 2, ..., n\}$ may be partitioned into different triplets (3-sets), so that in each triplet, one element is the sum of the other two.

   (Czech-Slovak-Poland Mathematical Olympiad 2007). Because the set $\{1, 2, ..., n\}$ may be partitioned into 3-sets, thus $3 \mid n$. Also into each triplet, we have the elements are a, b and $a+b$, hence an even sum. This means $2 \left| \frac{n(n+1)}{2} \right.$, or $4 \mid n(n+1)$. We get possible n are 3900 or 3903. We need a lemma:

   <u>Lemma</u>: Suppose the set $\{1, 2, ..., n\}$ contains the require partition when $n=k$, then so are the sets when $n=4k$ or $n=4k+3$. First if the set $\{1, 2, ..., k\}$ contains the required construction, then so is the set $\{2, 4, ..., 2k\}$. For $n=4k$, we need to consider the remaining elements
   $\{1, 3, 5, ..., 2k-1, 2k+1, 2k+2, ..., 4k-1, 4k\}$. These elements may be partitioned into k triplets $\{2j-1, 3k-j+1, 3k+j\}$ $(j=1, 2, ..., k)$. As shown by the columns:

$$\begin{pmatrix} 1 & 3 & 5 & ... & 2k-3 & 2k-1 \\ 3k & 3k-1 & 3k-2 & ... & 2k+2 & 2k+1 \\ 3k+1 & 3k+2 & 3k+3 & ... & 4k-1 & 4k \end{pmatrix}.$$

For $n=4k+3$, we need to consider the remaining elements $\{1,3,5,\ldots,2k-1,2k+1,2k+2,\ldots,4k+2,4k+3\}$. These elements may be partitioned into $k+1$ triplets $\{2j-1,3k-j+3,3k+j+2\}$ $(j=1,2,\ldots,k+1)$. As shown by the columns:

$$\begin{pmatrix} 1 & 3 & 5 & \ldots & 2k-1 & 2k+1 \\ 3k+2 & 3k+1 & 3k & \ldots & 2k+3 & 2k+2 \\ 3k+3 & 3k+4 & 3k+5 & \ldots & 4k+1 & 4k+3 \end{pmatrix}.$$

The lemma is then proved. Now we know for $n=3$, it works, we then move on as follows:

$$3 \xrightarrow{4\times3+3} 15 \xrightarrow{4\times15} 60 \xrightarrow{4\times60+3} 243 \xrightarrow{4\times243+3} 975.$$

For $n=975$, we get $4n=3900$, and $4n+3=3903$.

Remark: Ingenious construction.

6. Initially the edges of a complete graph with n vertices are colored by three different colors, with each color used at least once. Now assume suppose we choose at most k edges from this graph, and change the colors of these k edges to a particular color, so that the graph becomes connected using only the edges of this particular color. Determine the smallest possible value of k, so that no matter how the graph was initially colored, the goal may be attained.

(Korea Mathematical Olympiad 2015). $k_{\min}=\left[\dfrac{n}{3}\right]$. First show $k_{\min}\geq\left[\dfrac{n}{3}\right]$. Let the three colors be 1, 2, and 3, vertices of the n-complete graph are divided into three sets, A, B and C, with $|A|=|B|=\left[\dfrac{n}{3}\right]$. The edges are colored as follows. Edges within A are colored 1, within B colored 2, within C colored 3, edges connecting vertices A and B by color 2, B and C by colored 3, C and A by color 1. Suppose when we change the color of some edges and eventually color 1 is chosen to connect the graph, because the edges leaving B are initially colored 2 or 3, and among the edges of B are also colored 2, we need to change at least $\left[\dfrac{n}{3}\right]$ edges leaving B. Analogously if eventually we use color 2 or 3, we need to change the edges leaving C or B, with $|C| \geq \left[\frac{n}{3}\right]$. Hence $k_{\min} \geq \left[\frac{n}{3}\right]$.

Now we prove $k \leq \left[\frac{n}{3}\right]$. We proceed by induction. For $n = 3$, it is clear. Suppose by induction, the case for $n-1$ is valid, for $n \geq 4$. We now have a colored complete graph with n vertices. First case, if in the graph, there is a vertex connecting other vertices, with all three colors used, then by induction, we can change the colors of at most $\left[\frac{n-1}{3}\right]$ edges of the remaining $(n-1)$- complete graph to get a colored connected graph. Second case, suppose initially edges going out from any vertex are colored by at most 2 colors. Call A those vertices so that edges leaving them are colored only by color 1 or 2, B, colors 2 or 3, and C, colors 3 and 1. Without loss of generality, $|A| \leq |B| \leq |C|$. Then $|A| \leq \left[\frac{n}{3}\right]$. Now two sub-cases. First suppose B is an empty set. That means all edges are colored 1 or 3 (that suppose cannot happen). Say all edges are colored 1, then we are done. Suppose there are edges colored 1 or 3 exited from a certain vertices, then we may use the argument same as the first case and proceed by induction. Now if B is not that means all edges from B to C are actually colored by 3, hence $B \cup C$ can be connected by color 3. At this point, choose any point from B, and color all edges connecting this point and A by color 3. From $|A| \leq \left[\frac{n}{3}\right]$, we get the result.

Remark: Nice Induction and delicate analysis.

7. Given a positive integer n, determine the maximum number of lattice points in the plane a square of side length $n + \frac{1}{2n+1}$ may cover.

(Danube Mathematical Competition 2012). The required maximum is $(n+1)^2$. As the square $\left[-\frac{\varepsilon}{2}, n+\frac{\varepsilon}{2}\right] \times \left[-\frac{\varepsilon}{2}, n+\frac{\varepsilon}{2}\right]$, $0 \leq \varepsilon < 1$, covers exactly $(n+1)^2$ lattice points, (lattice points (p,q), $0 \leq p,q \leq n$, p, q integers.

Now we show any (closed) square of side length $n+\varepsilon, 0 \leq \varepsilon \leq \frac{1}{2n+1}$, covers at most $(n+1)^2$ lattice points. For $n = 1$, the diameter of a square of side length $1+\varepsilon$ is $(1+\varepsilon)\sqrt{2}$, wheras the diameter of any configuration of 5 lattice points (furthest distance between any two lattice points) is at least $\sqrt{5} > (1+\varepsilon)\sqrt{2}$ in the range $0 \le \varepsilon \le \frac{\sqrt{10}}{2} - 1$. Assume $n \ge 2$, consider the *convex hull* K of the lattice points covered by a square of side length $n+\varepsilon, 0 \le \varepsilon \le \frac{1}{2n+1}$. Clearly $K \le (n+\varepsilon)^2$, the area of the square, as it is convex and cover the convex hull. On the other hand, by *Pick*'s theorem, $K = m - \frac{k}{2} - 1$, where m is the number of lattice points covered by K, and k is the number of lattice points on the boundary of K. Therefore

$$m = \text{area } K + \frac{k}{2} + 1 \le (n+\varepsilon)^2 + \frac{k}{2} + 1.$$

We want to find an upper bound for k, as the perimeter of K does not exceed the perimeter of the square, which is $4(n+\varepsilon) \le 4n + \frac{4n}{2n+1} < 4n+1$, for $n \ge 2$. Since the distance between two lattice points is at least 1, it follows that $k \le 4n$. Consequently $m \le (n+\varepsilon)^2 + 2n + 1 = (n+1)^2 + 2n\varepsilon + \varepsilon^2 < (n+1)^2 + 1$, in the slight range $0 \le \varepsilon < \sqrt{n^2+1} - n$, and we are done.

Remark: Combinatorial Geometry, use the concepts of the convex hull, and Pick's theorem.

8. Let $A_1, A_2, ..., A_n$ be finite sets satisfying

$$|A_i \cap A_{i+1}| > \frac{n-2}{n-1}|A_{i+1}|, i = 1, 2, ..., n \; (A_{n+1} = A_1).$$

Show that $\bigcap_{i=1}^{n} A_i \ne \Phi$.

(China Selection Test 2007). Without loss of generality, assume A_1 contains the largest number of elements as compared with others. Let $B_i = A_i \cap A_{i+1}, i = 1, 2, ..., n$. Then

$$|A_n| \ge |B_n \cup B_{n-1}| = |B_n| + |B_{n-1}| - |B_{n-1} \cap B_n| > \frac{n-2}{n-1}|A_n| + \frac{n-2}{n-1}|A_1| - |B_{n-1} \cap B_n|.$$

Hence $|B_{n-1} \cap B_n| > \frac{n-2}{n-1}|A_1| - \frac{1}{n-1}|A_n| \ge \frac{n-3}{n-1}|A_1|$. Thus

$|A_{n-1} \cap A_n \cap A_1| > \frac{n-3}{n-1}|A_1|$. Now let $C = A_{n-1} \cap A_n \cap A_1$, then $A_{n-1} \supset C \cup B_{n-2}$. So

$|A_{n-1}| \ge |B_{n-2} \cup C| = |B_{n-2}| + |C| - |B_{n-2} \cap C| > \frac{n-2}{n-1}|A_{n-1}| + \frac{n-3}{n-1}|A_1| - |B_{n-2} \cap C|$. Thus

$\left|B_{n-2} \cap C\right| > \frac{n-3}{n-1}\left|A_1\right| - \frac{1}{n-1}\left|A_{n-1}\right| \geq \frac{n-4}{n-1}\left|A_1\right|$. This implies

$\left|A_{n-2} \cap A_{n-1} \cap A_n \cap A_1\right| > \frac{n-4}{n-1}\left|A_1\right|$. By induction, get

$\left|A_{n-k} \cap A_{n-k+1} \cap \ldots A_n \cap A_1\right| > \frac{n-k-2}{n-1}\left|A_1\right|, k = 1, 2, \ldots, n-2$. Hence

$\left|A_2 \cap A_3 \cap \ldots A_n \cap A_1\right| > 0$.

Remark: Extremal Principle, Inclusion-Exclusion, Induction, technical but not too hard.

9. We are given two square sheets of paper with area 2003. Suppose we divide each of these papers into 2003 polygons, each of area 1. (The divisions of the two pieces of these papers may be distinct. Then we place the two sheets of paper on top of each other. Show that we can place 2003 pins on the pieces of papers so that all 4006 polygons have been pieced.

(Kazakhstan 2003). For each subset W of the 2003 polygons on one sheet of papers, we want to show that there are at least $|W|$ polygons on the other sheet of paper that touch this subset. Indeed suppose not, there are fewer than $|W|$ polygons touching it, i.e. the *neighborhood* $N(W)$ of W, that means the polygons of W are smaller than those of $N(W)$ because they occupy the same area. As the area of W is $|W|$, the area of $N(W)$ is $|N(W)|$, hence $|W| \leq |N(W)|$. Now we define a bipartite graph connecting X and Y, where X are the set of top polygons, and Y the bottom polygons, connecting two polygons if and only if they have non-trivial intersection by a pin. Now by the *Hall***'s** *marriage theorem,* there is a matching between 2003 pairs of top and bottom polygons. OK.

Remark: (Hall's marriage theorem). Let G be a bipartite graph with bipartite sets X and Y. Then there exists a matching that covers X if and only if for each subset W of X, $|W| \leq |N(W)|$, where $N(W)$ are those vertices in Y which are adjacent to any vertex in X.

Section C

2. Prove that: for any integer $n > 16$, there exists a set S consisting of n integers, with the following property: if S has a subset A, such that for any $a, a' \in A,\ a \neq a',\quad a + a' \notin S$, then $|A| \leq 4\sqrt{n}$.

(Chinese Selection Test 2008). Let k be a parameter, $1 < k < n$, let $n = kq + r, 0 \leq r < k$ (note $q = \left[\frac{n}{k}\right]$). For $i = 1, 2, ..., k$, define $S_i = \{2^{i-1}m \mid q \leq m \leq 2q - 1\}$, then $|S_i| = q$, and clearly for $1 \leq i < j \leq k$, $S_i \cap S_j = \phi$. Take S_0 be a set consisting of r integers, so that for any $1 \leq i \leq k$, $S_0 \cap S_i = \phi$. Now take $S = \bigcup_{i=0}^{k} S_i$. Then $|S| = kq + r = n$. We now prove S has the required properties.

Clearly S has the property A, such that for any $a, a' \in A,\ a \neq a',\quad a + a' \notin S$, (just let A small enough). Now we prove that for any such set A, for any $1 \leq i \leq k - 1$, $|S_i \cap A| \leq 2$. Indeed if there is an i, $(1 \leq i \leq k - 1)$, $|S_i \cap A| \geq 3$. Let $2^{i-1}m_1, 2^{i-1}m_2, 2^{i-1}m_3$ be 3 members of $S_i \cap A$, where $q \leq m_1, m_2, m_3 \leq 2q - 1$, then two of m_1, m_2, m_3 are of same parity, say m_1 and m_2, say $m_1 + m_2 = 2m$, so $q \leq m \leq 2q - 1$, hence $2^{i-1}m_1 + 2^{i-1}m_2 = 2^i m \in S_{i+1} \subseteq S$, violating the condition of A, hence $|S_i \cap A| \leq 2$.

From this, and using $r \leq k - 1$, we get

$$|A| \leq |S_0| + |S_k| + 2(k-1) = r + q + 2(k-1) < 3k + \left[\frac{n}{k}\right] - 3 \leq 3k + \frac{n}{k} - 3.$$

Now take $k = \left[\sqrt{\frac{n}{3}}\right] + 1$, with $n > 16$, get $1 < k < n$, hence $|A| < 2\sqrt{3n} < 4\sqrt{n}$.

OK.

Remark: Ingenious construction. First find a set satisfying the first condition, then use the parameter to get the second condition.